



CPSC 100

Computational Thinking

Data Visualization

Instructor: Firas Moosvi
Department of Computer Science
University of British Columbia

Course Admin



Here is an example:

- A group of four students (Nancy, Robin, and Vecna) are called **STRANGER_THINGS** and they want to aim for an A+ project
- **STRANGER_THINGS** chooses from one of the **5 broad themes**: "*Financial and Environmental impacts of Technology*"
- The group has a main **area of inquiry**: "*Electric cars and Self Driving Cars*" and each member has the following interests/angles/slants/directions:
 - **Vecna** wants to investigate how AI is affecting car manufacturing companies particularly from a financial and economic side. Vecna plans to explore this concept using the following course topics:
 - AI (1)
 - Ethics and Computer Science (2)
 - Social Implications and Computing (3)
 - Data Storage (4)
 - Data Security (5)
 - **Max** wants to explore the environmental and social/ethical impacts of electric vehicles. Vecna plans to explore this concept using the following course topics:
 - AI
 - Ethics and Computer Science
 - Social Implications and Computing
 - Data Privacy (6)
 - Nancy wants to explore the financial and environmental impacts of companies like Waymo using self-driving cars for ride-sharing. Vecna plans to explore this concept using the following course topics:
 - AI
 - Algorithms (7)
 - Data Representation (8)
 - Data Mining (9)
 - Data Security



Learning Goals (Part 1)

After this **today's lecture**, you should be able to:

- Define infographics and their role in conveying information effectively.
 - Understand the difference between infographic vs. visualization
- Recognize, define and apply high-level principles of infographic design
- Identify strengths and weaknesses in infographic designs based on high-level principles.



Learning Goals (Part 2)

After this **today's lecture**, you should be able to:

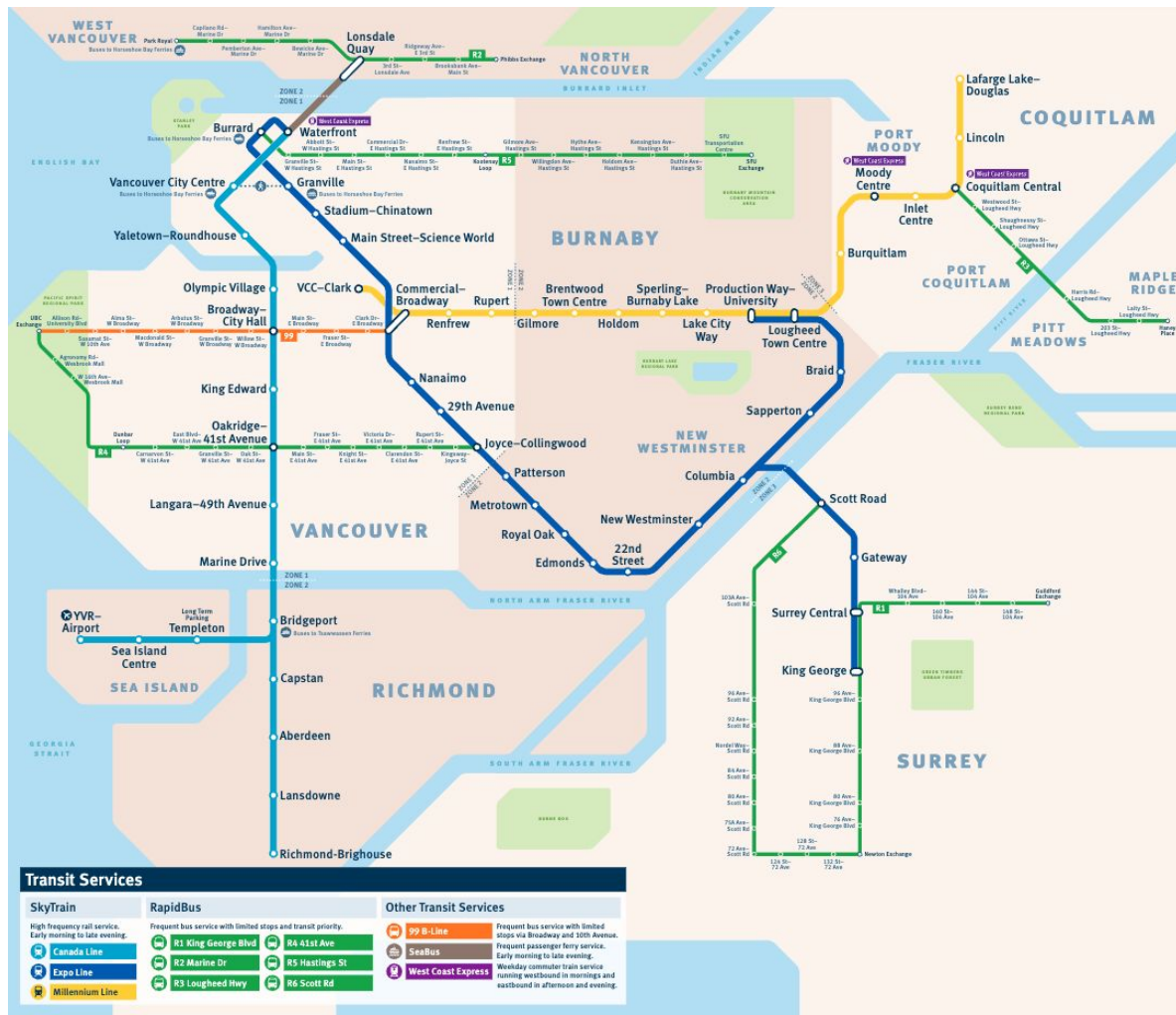
- Explain how different chart types are used to represent and compare data.
- Apply knowledge of data types (categorical, quantitative) to select suitable visualizations to convey information.
- Explain how different visualizations contribute to an effective data story.

After watching the **take-home video**, you should be able to:

- Differentiate between effective and ineffective use of **visualization channels** in data representation.

Infographics

Vancouver





What are Infographics?

"Information graphics or infographics are **visual representations of data**, information, or knowledge intended to **present the idea quickly succinctly, and clearly.**" (Wikipedia)

- Static representation that conveys a specific message
- Typically includes graphics and stats (but it doesn't have to)
- Images used do not necessarily have to encode data



Infographics vs. Visualization

- An infographic is a **static representation** that conveys a specific message. It typically includes graphics and stats but doesn't have to.
- Visualization can be an **interactive or static representation** that primarily uses visual marks to encode data.

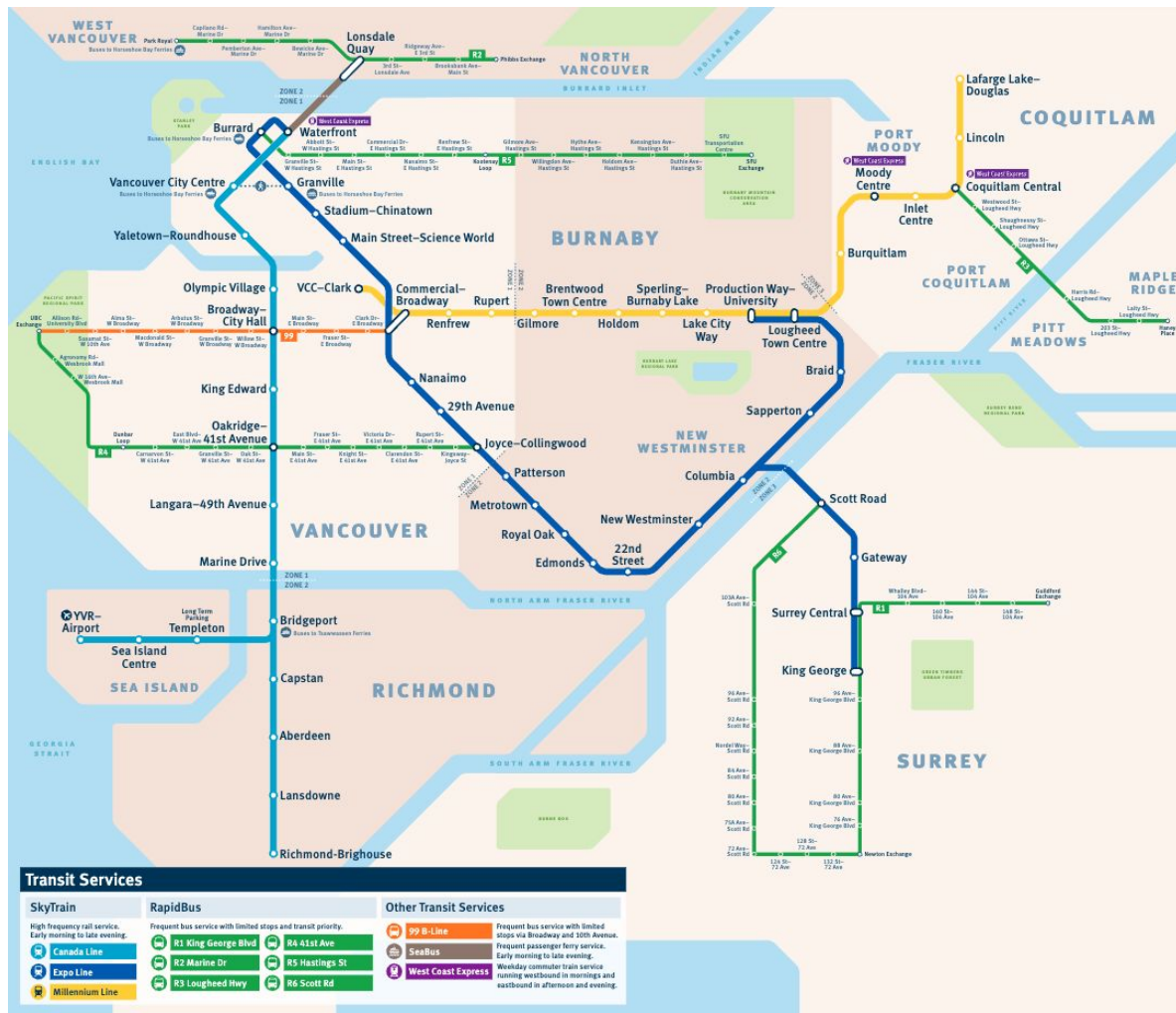


Why Infographics?

- Improve Comprehension and Retention
 - mix of text, visuals, and icons to break down complex ideas
- Enhance Engagement
 - attracts more attention than plain text
- Simplify Data and Statistics
 - make numbers more understandable*
 - * = simplification should not compromise the accuracy of the data
- Increase Information Accessibility
 - bridge language and literacy gaps by using icons, symbols, and structured layouts

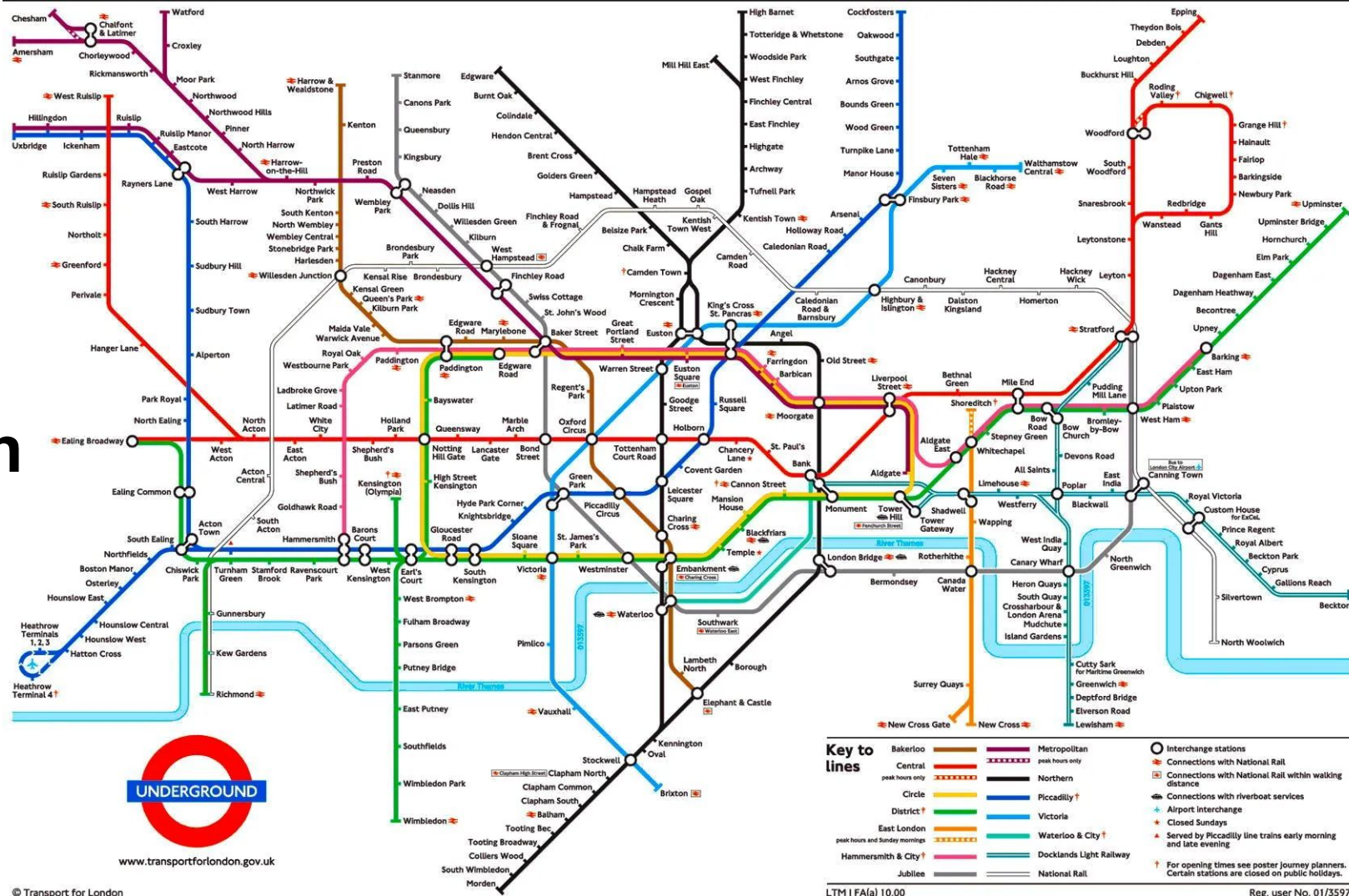
Examples: Transit Maps

Vancouver





London

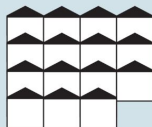




Other Examples

UBC Library Snapshot

UBC Library advances research, learning and teaching excellence by connecting communities within and beyond UBC to the world's knowledge. The Library, a high-ranking member of the Association of Research Libraries (ARL), is the largest library in British Columbia and provides access to expanding digital resources and houses an on-site digitization centre. For more information, visit library.ubc.ca.



15 branches across
2 campuses



315 full-time staff

- 88 librarians
- 184 management & support staff
- 43 student employees

Rankings

- **14 out of 115** university libraries in the Association of Research Libraries (ARL)
- **2nd** among Canadian academic libraries (ARL)
- cIRcle, UBC's information repository, **ranks 2nd in Canada** and 44th globally among 1,650 repositories.

COLLECTIONS

› More than **7.4m** volumes

› More than **1.8m** e-books

› **330,000*** e-journals

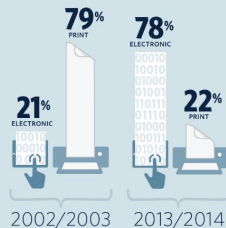
› **500,000*** items in

locally produced **digital collections**

More than
5 million e-book downloads
8 million e-journal downloads

LIBRARY RESOURCES

Expenditures shifting
from **print** to **electronic**



on-campus **3.8M⁺**

visits (JAN-DEC 2013)

library.ubc.ca **11.1M⁺**

UBC Library on SOCIAL MEDIA

Twitter **4,8K**

16 accounts

Facebook **9,9K**

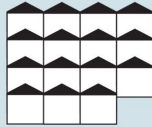


Librarians provided
1,768 instructionals
to more than **41,427**
participants
and answered
reference questions

- 54,648 in-person
- 10,756 online

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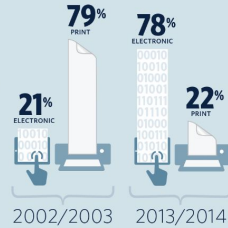
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Company: Benevity

Principles for Infographics



Principles for Infographics

- **Simplicity**
- **Consistency**
- **Visibility**
- **Navigability** (structure)
- **Suitability**



Principles for Infographics

- **Simplicity** – minimal text, clear message, avoid clutter
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Principles for Infographics

- **Simplicity** – minimal text, clear message, avoid clutter
- **Consistency** – layout and design elements should be consistent.
 - 2 to 3 font sizes, colour scheme
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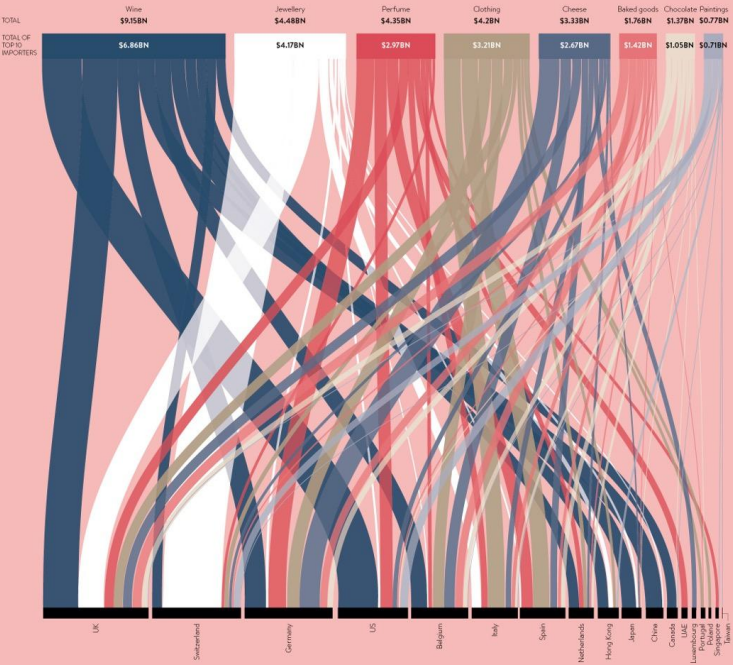
Principles in Action

Made in France

France exported \$572 billion of goods in 2015, making it the sixth largest exporter in the world. While aircraft, cars and medicines are the country's highest-valued goods, there are many other exports to varied markets

FRENCH EXPORTS AND TOP 10 IMPORTERS

Source: Trade Map/UN Comtrade 2016



ALTERNATIVE EXPORTS AND MOST POPULAR MARKETS

Source: Trade Map 2016

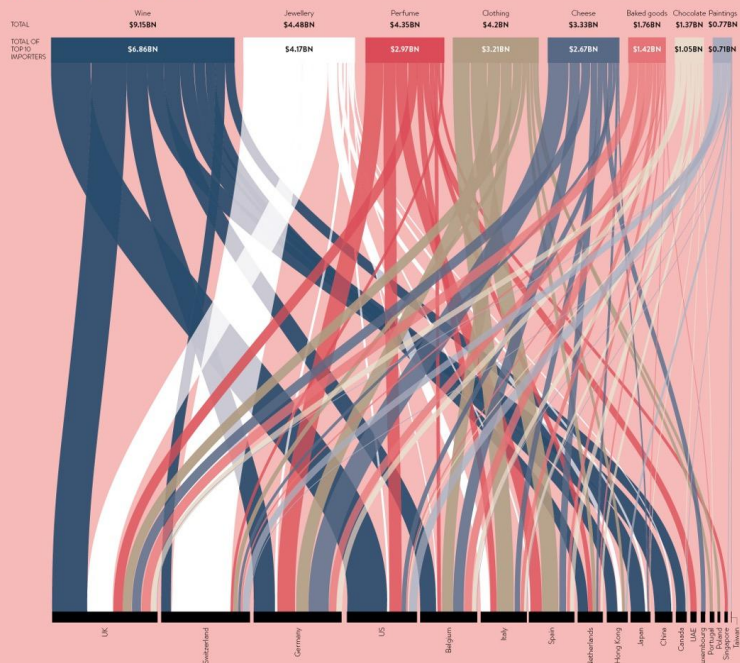


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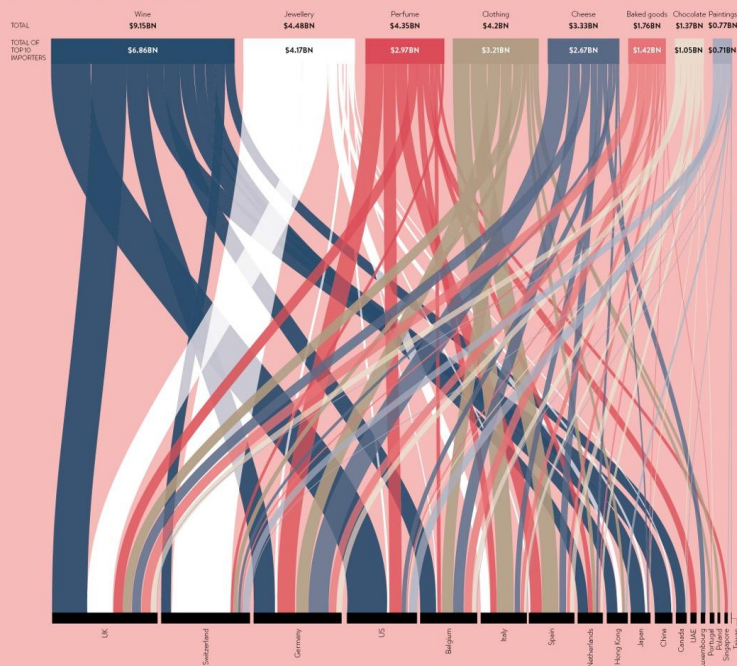


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✗ **Simplicity**

use of a sankey diagram does not help

✗ **Visibility**

text at the bottom is very hard to read

✗ **Suitability**

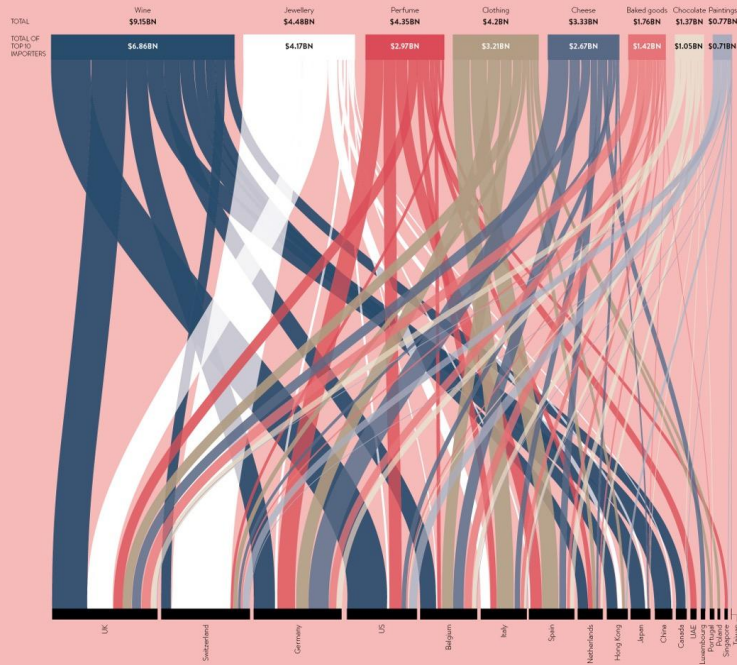
what is the focus of the infographic?

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✗ Simplicity

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✗ Suitability

what is the focus of the infographic?

✓ Navigability

top to bottom

✓ Consistency

colour theme

Infographics vs. Visualization

- An infographic is not a type of visualization, as it could just be made up of words and images. (e.g., 70% of children like chocolate ice cream with a picture of an ice cream cone).
- The images in an infographic do not necessarily have to encode data. So you can have infographic that has an emoji, an image of an animal, food, but not encode quantitative data with the images (e.g. no bar charts).

Activity



Activity

- On the next few slides are some examples of good (denoted by ) and bad (denoted by ) infographics.
- As part of this activity, try to make sense of the infographic, and determine which principle(s) the infographic violate or adhere to.
 - *Hint 1: some infographics may violate/adhere to more than 1 principle.*
 - *Hint 2: "good" infographics may also be violating some principles and vice-versa, a "bad" infographic may be adhere to some principles.*

CHEMICALS IN CIGARETTE SMOKE (SELECTED FROM OVER 4,000 TOTAL)

- ACETANISOLE
- ACETIC ACID
- ACETONE
- ACETOPHENONE
- ALFALEA EXTRACT
- ALLYL HEXANOATE
- ALLYL IONONE
- ALMOND BITTER OIL
- ALUMINUM
- AMMONIA
- AMMONIUM BICARBONATE
- AMMONIUM SULFIDE
- ARSENIC
- BENZALDEHYDE GLYCERYL
- BERYLLIUM (METAL)
- BUTANOL
- BUTANE
- CAFFEINE
- CALCIUM CARBONATE
- CARBON DIOXIDE
- CARBON MONOXIDE
- CITRONELLA OIL
- CHLOROFORM
- COCONUT OIL
- COFFEE
- CRIMINALBENYDE
- CYANIDE
- CYSTEINE
- DANOLILION ROOT EXTRACT
- DAVANA OIL
- DDT/DIELDRIN
- EUCALYPTOL
- FARNESOL
- FENCONE
- FENNEL SWEET OIL
- FIG JUICE
- FORMALDEHYDE
- LACTIC ACID
- MACE POWDER EXTRACT
- MENTHOL
- PATCHOULI OIL
- RUM ETHER
- SILICON
- SKATOLE
- SMOKE FLAVOR
- SNAKERROOT OIL
- SODIUM ACETATE
- SOLANONE
- TOBACCO EXTRACTS
- UREA



NAIL POLISH REMOVER



BATHROOM CLEANER



RAT POISON



CAR EXHAUST FUMES



PEANUT BUTTER



INSECTICIDE



PRESERVATIVE OF MEAT FUMES



BOTTLE CAP



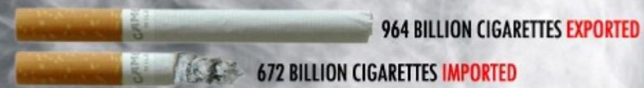
URINE

SMOKY TRAILS MAJOR SMUGGLING ROUTES



WHERE DID ALL THE CIGARETTES GO?

In 2000, there was a disparity of 292 billion cigarettes between worldwide exports and imports. In other words, nearly 1/3 of the world's total cigarette production simply disappeared, at least on paper. The global tobacco black market is booming and big tobacco companies like Phillip Morris and British American Tobacco are leading the charge. Lured by untapped markets in developing nations like India, Colombia and Bangladesh, and desperate to avoid soaring import taxes in countries like America, big tobacco is fueling a big business, and all of it is off the books.



WHO SMOKES & HOW MUCH?

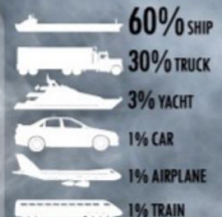
The World Health Organization estimates about 1.25 billion smokers in the world, roughly 1/3 of all people over the age of 15. With 1/5 of the world's population, China smokes 1/3 of its cigarettes. 61% of Chinese men are smokers, a world high.



SSSMOKIN! THE GLOBAL TOBACCO TRADE

1000
PACKS
OF MARLBOROS

111 DAYS
OF LIFE



1 LEATHER JACKET

Buying 1000 packs of Marlboro cigarettes will give you enough "Marlboro Miles" to win a free leather jacket. If each smoked cigarette takes 8 minutes of your life, that jacket will cost you 111 days. Enjoy!



TOOLS OF THE TRADE

BASED ON SMUGGLERS CAUGHT IN 2000 IN THE EUROPEAN UNION

THE THREE-TRILLION-DOLLAR WAR ITS COST IN TEN STEPS

In 2003, Secretary of Defense Donald Rumsfeld estimated that a war with Iraq would cost \$60 billion. Five years later, the cost of Iraq War operations is more than 10 times that figure. By the time the United States leaves Iraq, the estimated cost of war will be more than \$3 trillion.





The Big Cheese

Papa John's founder John Schnatter has been the dominant presence at his company since its founding—even after turning over the CEO role to a rotating cast of executives



John Schnatter
1990–January 2005

Steve Ritchie
January 2018 – Present

Jude Thompson
Co-CEO April 2010 – April 2011

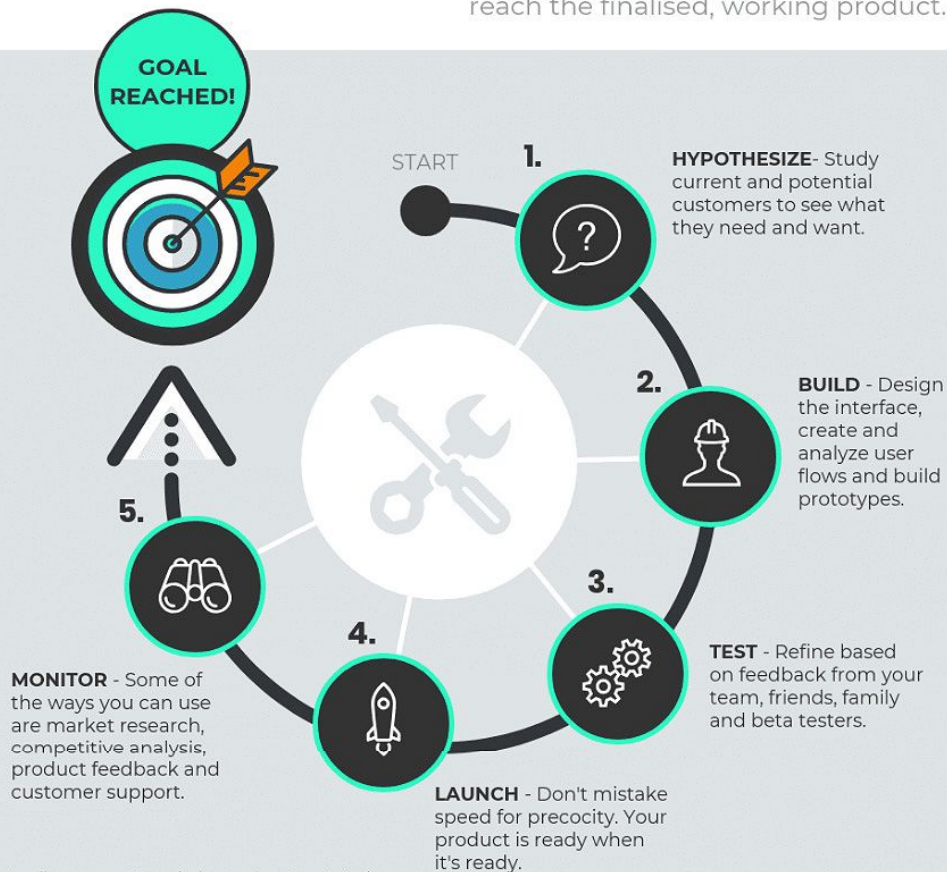
John Schnatter
December 2008 – December 2017

Nigel Travis
January 2005 – December 2008



STEPS IN PRODUCT DESIGN

Product design involves many steps in order to reach the finalised, working product.





5 Tips

To Keep Your Chin Up

1



Do something impulsive.

Do something impulsive that you haven't planned every day. It's better to have no plan so we can seize the opportunities that may arise.

2



Have rituals.

We are less who we are than what we do. Do 3 things that you love every day. As a result, feeling the gratitude will help you better sleep. Better sleep helps to be in a better mood. A better mood helps to make better decisions.

3



Exercise at least 10 minutes a day.

Exercising has an influence on your brain, on your mood, on your ability to reflect and on your health.

4



Take breaks.

Prevent burnouts by stopping what you are doing and do something else. Create a different atmosphere, add some novelties in your daily routine.

5



Learn something new.

Learning helps to create new connections in your brain and to come up with new ideas and new opportunities.



COVID-19 And The Refugee Crisis Timeline

March 11, 2020

World Health Organization declares a **global pandemic**



April 4, 2020

1 million confirmed cases of COVID-19 worldwide



May 14, 2020

COVID-19 case found in Bangladesh Rohingya refugee camps



June 10, 2020

Asylum applications reach lowest level in Europe



March 18, 2020

All new refugee admissions to the U.S. are **suspended**



April 10, 2020

Refugee settlements begin building COVID-19 **isolation centers**



May 25, 2020

400 people attempt to flee Libya, **317** are caught and returned



Visualization Zoo



Visualization Zoo

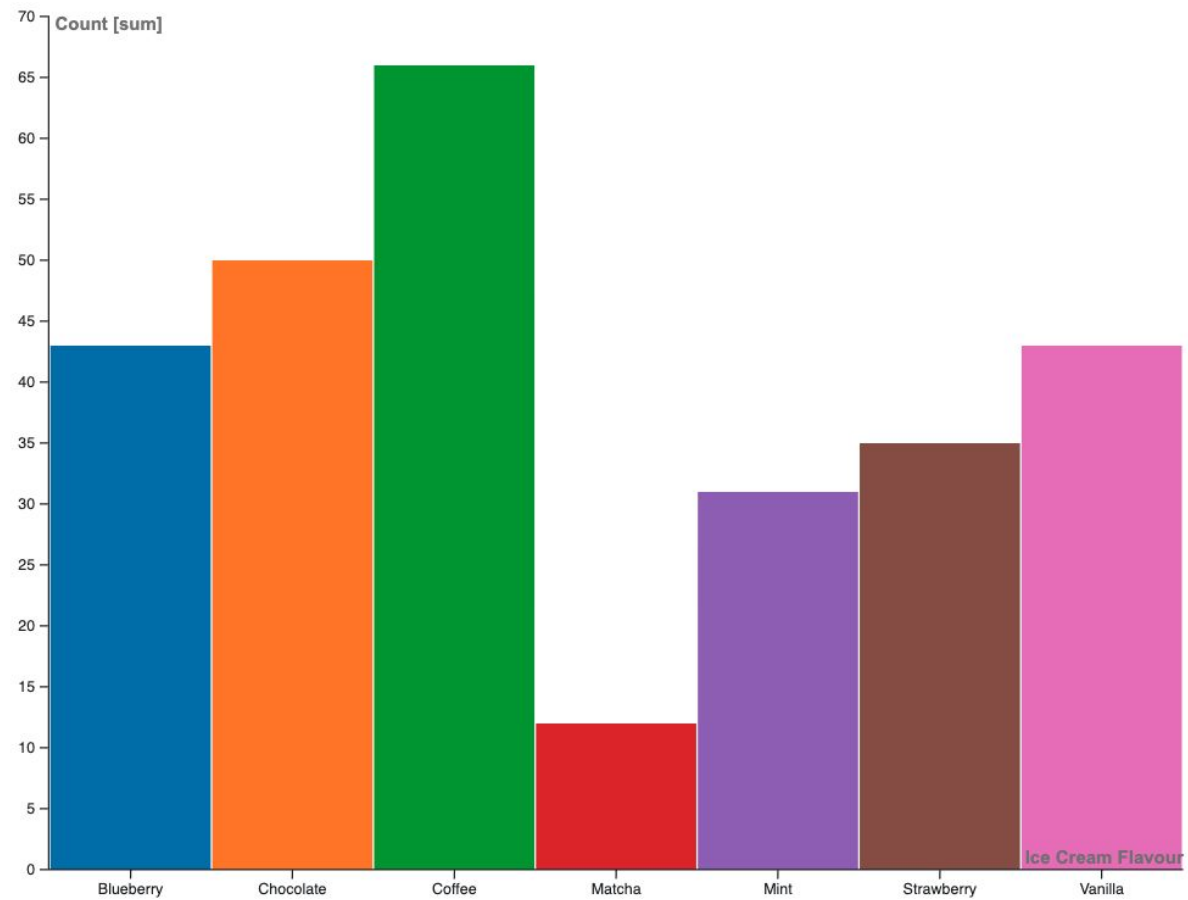
- **Comparing Categories**
 - Bar Chart: Bar, Stacked Bar, Grouped Bar
 - Pie Chart
- **Discovering Temporal Trends**
 - Line, Multi-Line, Area Charts
- **Multiple Attributes**
 - Radar Chart, Bubble Charts, Parallel Coordinates
- **Visualizing Relationships and Hierarchy**
 - Tree Maps, Scatterplots, Tree Diagrams, Network Graph

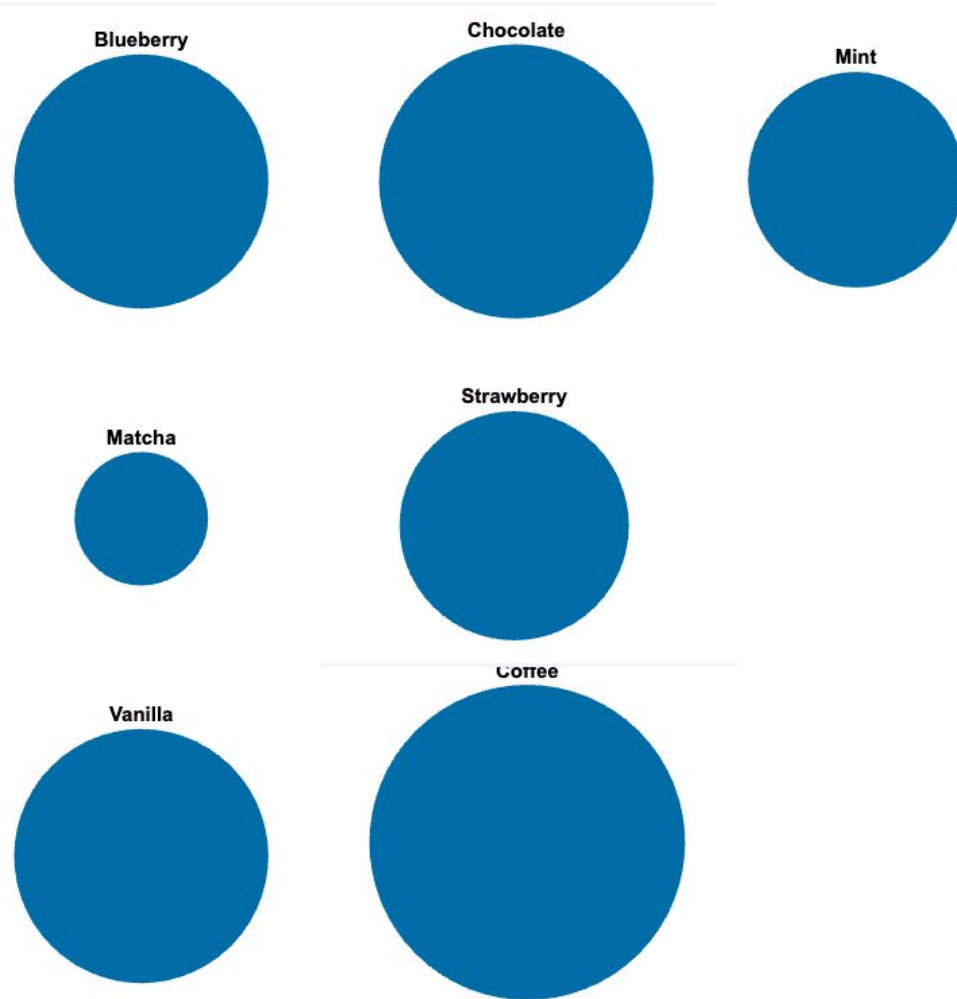
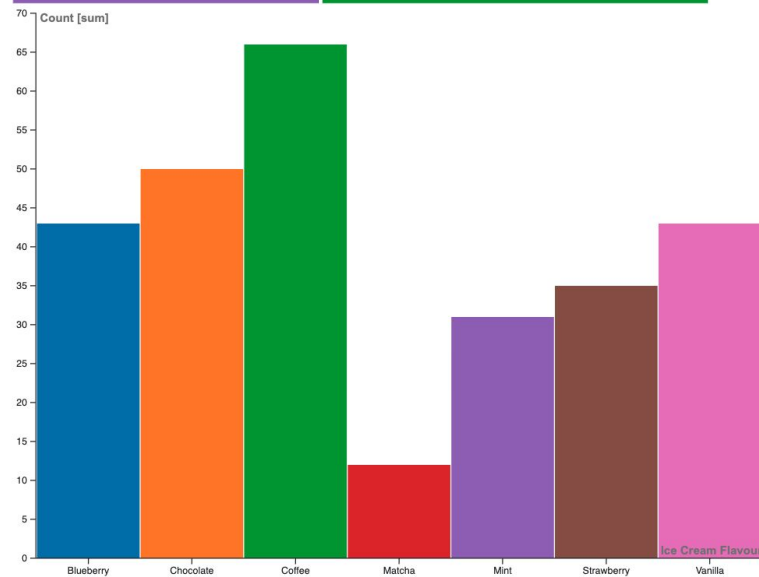
Activity

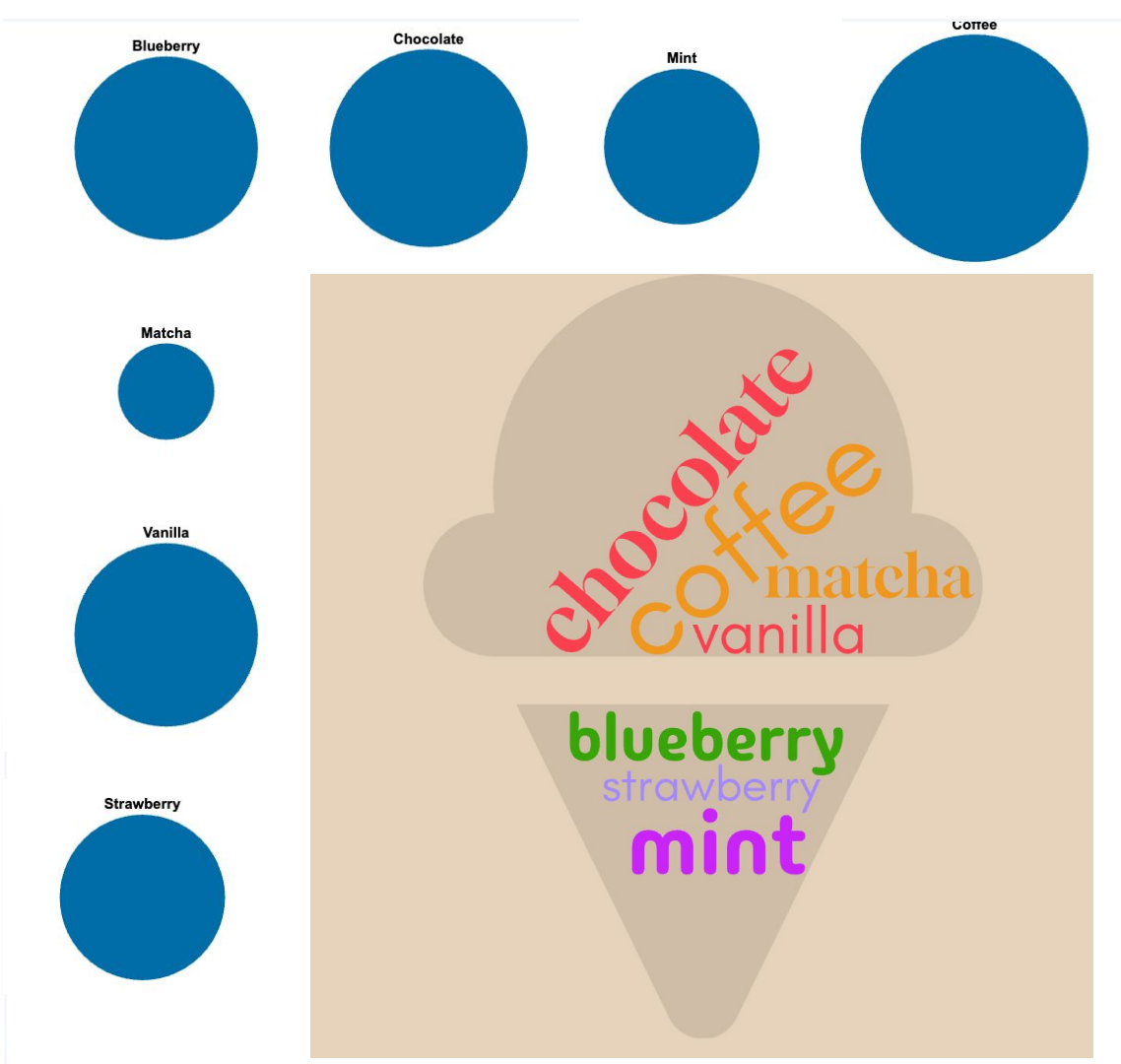
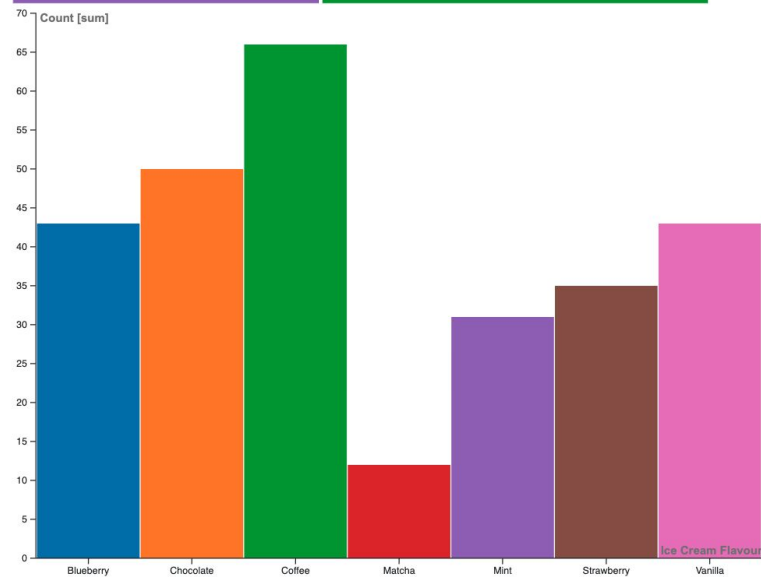
- Visualize the dataset below in at least 5 different ways.
- Sketch on paper/PrairieLearn
OR use a tool

Ice Cream Flavour	Count
Chocolate	50
Vanilla	43
Mint	31
Strawberry	35
Matcha	12
Blueberry	43
Coffee	66









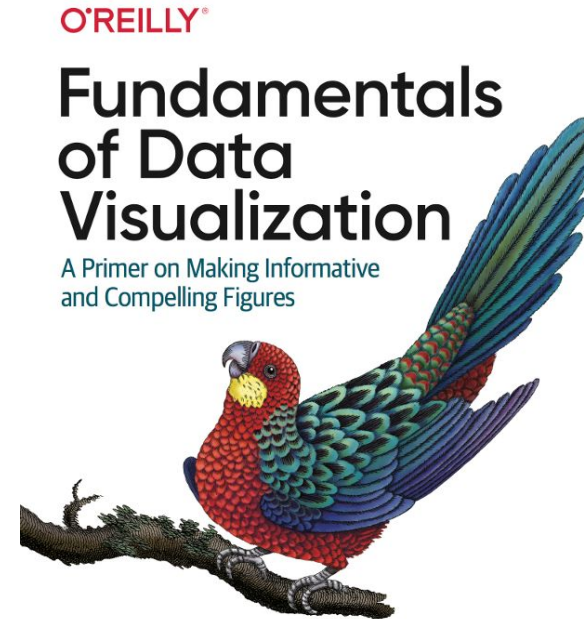
Principles of Effective Visualizations

Principle	Definition	Examples
• Proportional Ink	The amount of ink used to indicate a value should be proportional to the value itself.	Truncating the y-axis on a bar chart to exaggerate the difference between bars violates the principle of proportional ink.
• Data:ink ratio	Remove distracting visual elements to focus attention on the data	Lighten line weights, remove backgrounds, never use 3D or special effects, remove avoid unnecessary/redundant labels.
• Labels & legends	Use axes labels and titles to highlight/communicate data	Never leave your data column names as axes labels! Generally good to add a title.
• Overplotting	With large datasets, points overlap, resulting in large clouds of data	To fix overplotting, could plot just a sample subset of the data, use alpha, and use smaller points. Or, jitter - but check if appropriate!
• Visualization choice	Must be informed by the data you have, the research question being asked and the audience that cares.	Pick the simplest plot that best shows most/all of the data needed to answer the research question. If you only have summary statistics, cannot show distributions. Tailor the visualization to your audience (within reason) but don't dumb it down.
• Colour & Accessibility	Colour can be used to encode information or for aesthetics/style/design. However, colour can also be distracting if used inappropriately or poorly.	Choose a perceptually uniform colour palette; can be sequential or diverging for quantitative data. Opt for colour-blind friendly palettes. Categorical data can use qualitative colour schemes.



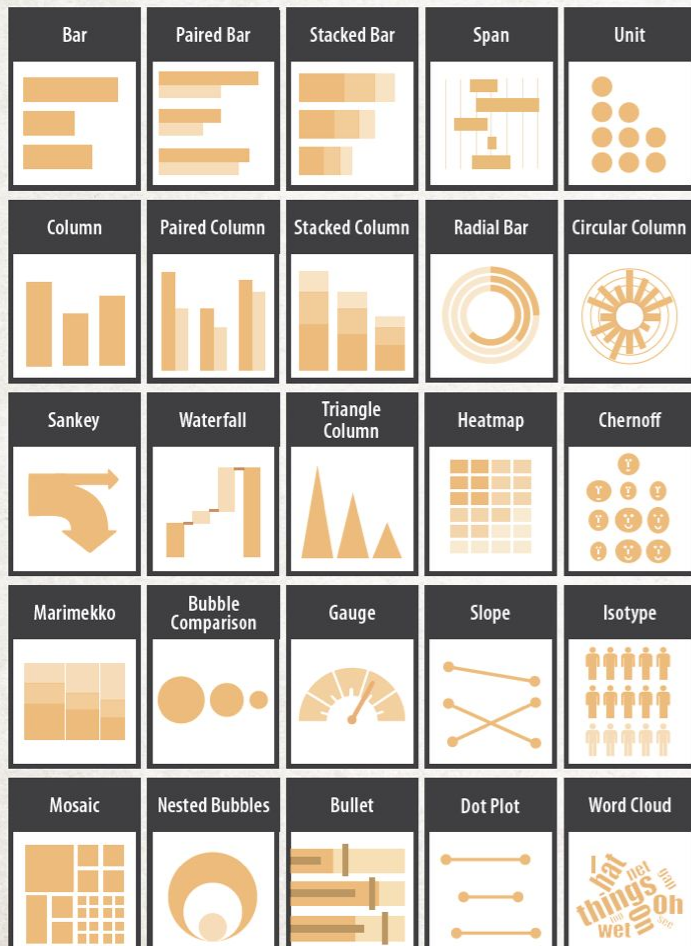
Welcome to the wonderful world of Data Visualization!

- Highly recommend this [free online book](#) as a reference!



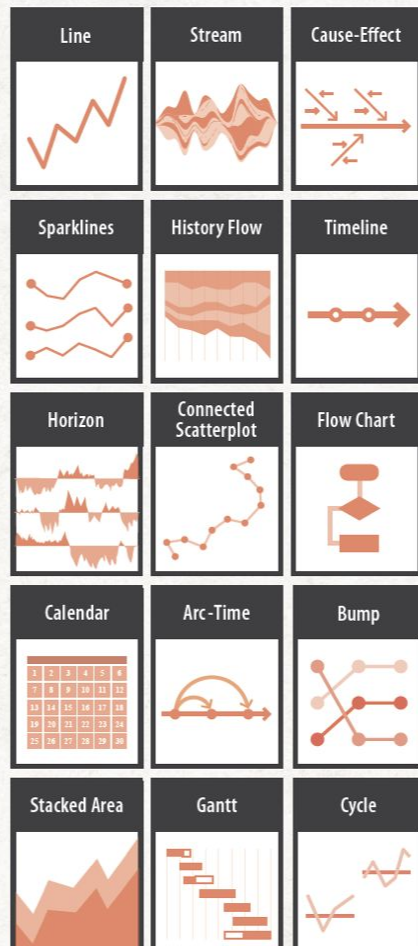
COMPARING CATEGORIES

Compare values across categories



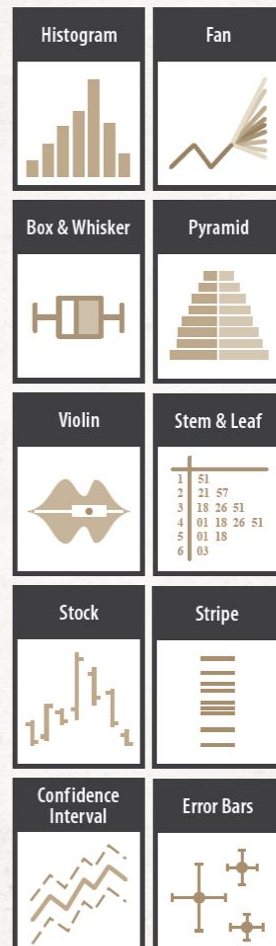
TIME

Track changes over time



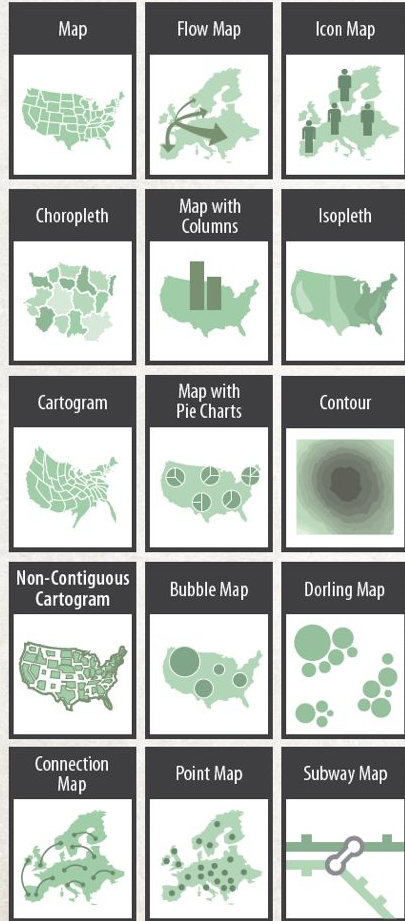
DISTRIBUTION

Representation of the distribution of data



GEOSPATIAL

Relates data to its geography



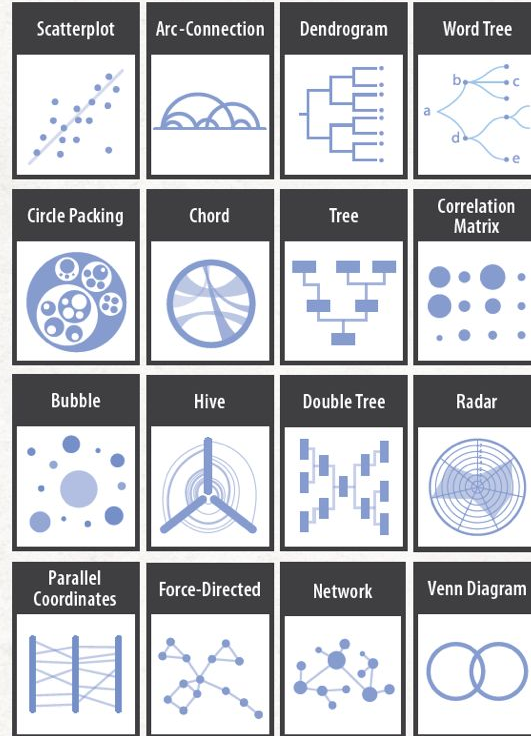
PART-TO-WHOLE

Relates the part of a variable to its total



RELATIONSHIP

Illustrates correlations or relationships between variables





Data Types

- **Categorical Data**
 - Represents distinct groups or labels (e.g., cities, ice cream flavors, departments)
 - Typically non-numeric, though can be encoded as such
- **Numerical (Quantitative) Data**
 - Measures quantities
 - Can be discrete (counts) or continuous (time, temperature)
- **Multivariate Data**
 - Contains more than two variables per observation

The Data Visualisation Catalogue

Search by Function

View by List



Arc Diagram



Area Graph



Bar Chart



Box & Whisker Plot



Brainstorm



Bubble Chart



Bubble Map



Bullet Graph



Calendar



Candlestick Chart



Chord Diagram



Choropleth Map



Circle Packing



Connection Map



Density Plot



Donut Chart



Dot Map



Dot Matrix Chart



<https://datavizcatalogue.com/index.html>

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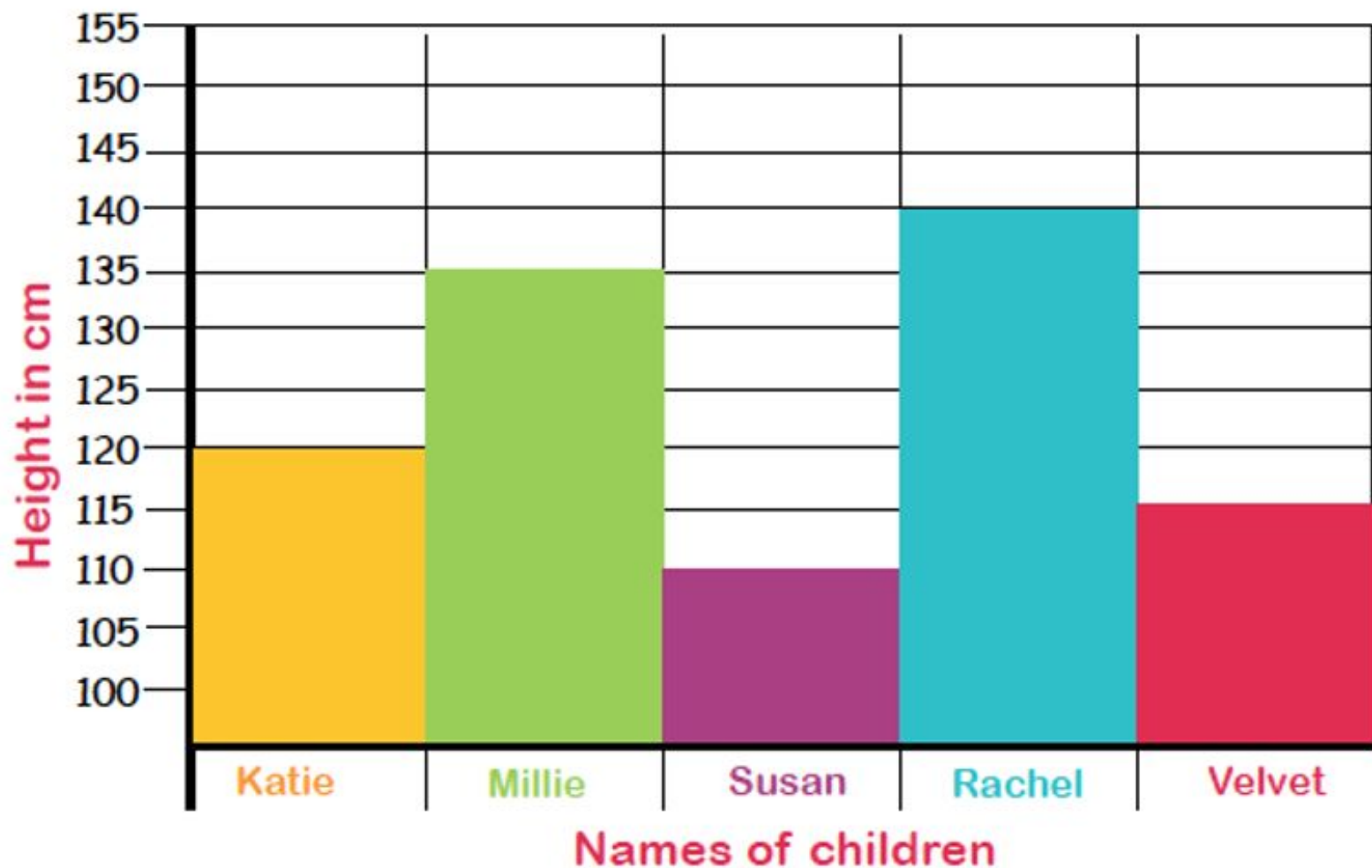
<https://datavizproject.com/#>





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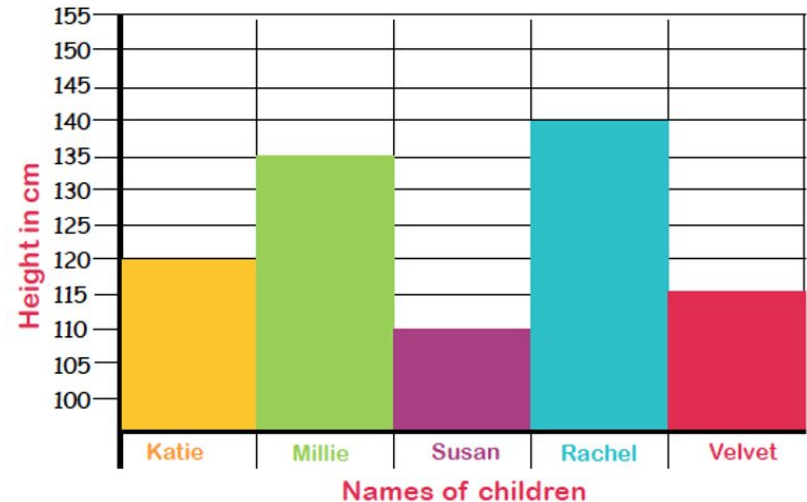


Bar Chart

- Use for comparing discrete categorical data (e.g., product sales, survey responses)
 - Answers the question of "how many?" in each category.
- it does not display **continuous developments** over an interval
- Easily readable and interpretable.

Data type:

Categorical (x-axis) + Numeric (y-axis)

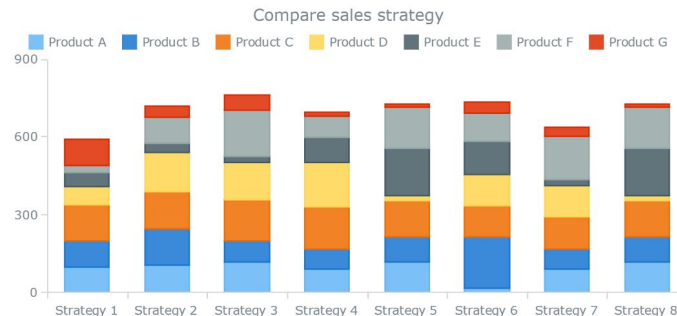


Compare sales strategy



Stacked Bar Chart

- Good for proportions, parts to a whole, comparisons
 - Shows sub-category composition within each main category.
- **Limitation(s):**
 - Harder to read the more segments each bar has.
 - Comparing segments to each other is difficult, as they're not aligned on a common baseline.
- **Data type:**
 - Categorical with grouped sub-categories

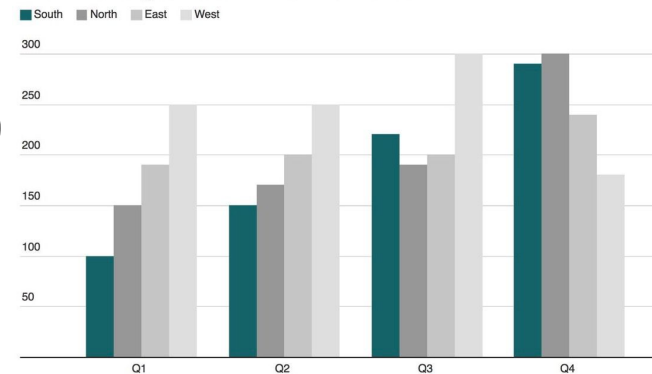




Multi-Set or Grouped Bar Chart

- Good for distribution, relationships and comparisons
 - Easier than stacked bar for subgroup comparison
- Limitation(s):
 - Becomes unreadable with too many groups
- Data type:
 - Categorical (grouping + sub-grouping)

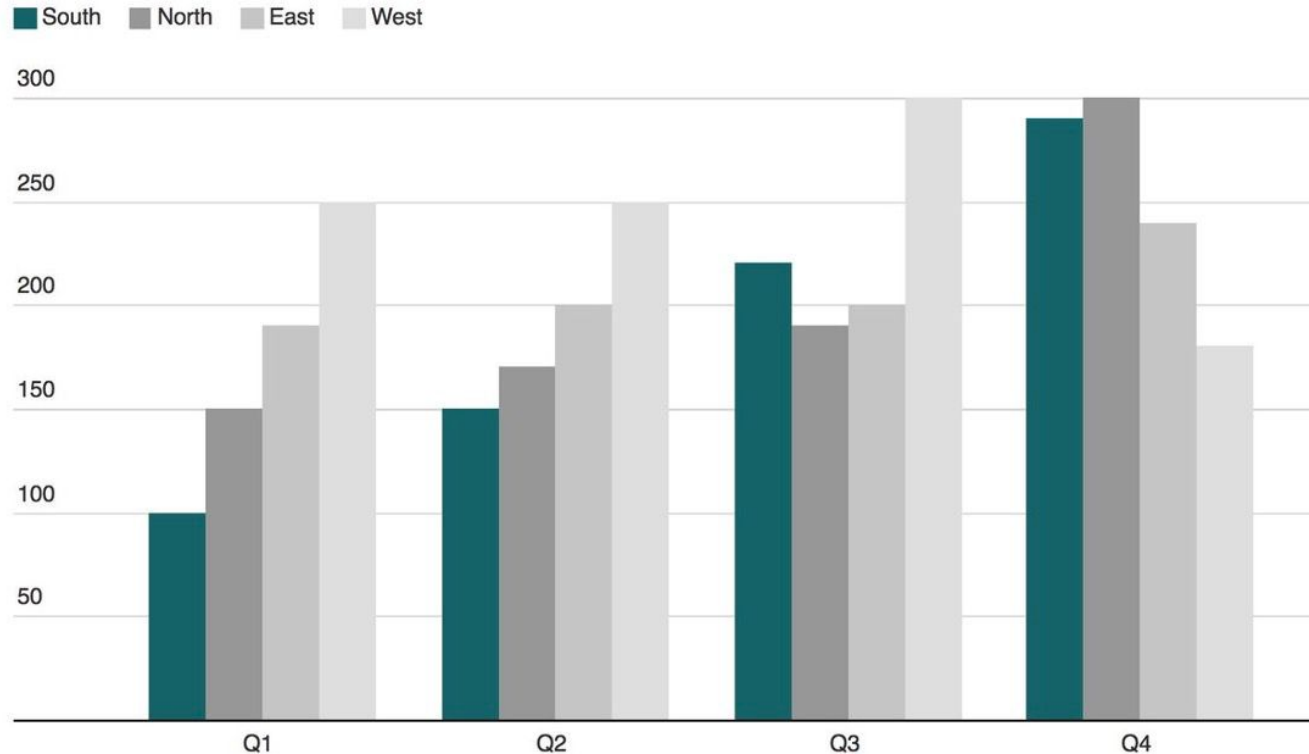
The South Region Exhibits Greatest Growth!





Multi-Set or Grouped Bar Chart

The South Region Exhibits Greatest Growth!





Pie Chart

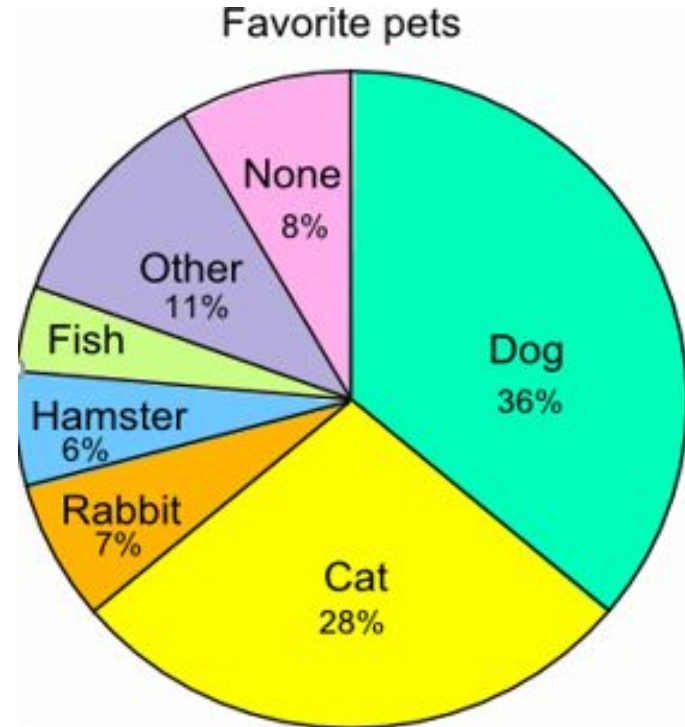
- Show part-to-whole relationships
 - Quick visual impression of proportions

Limitation(s):

- Poor for precise comparison
- Difficult to read with more than ~5 slices

Data type:

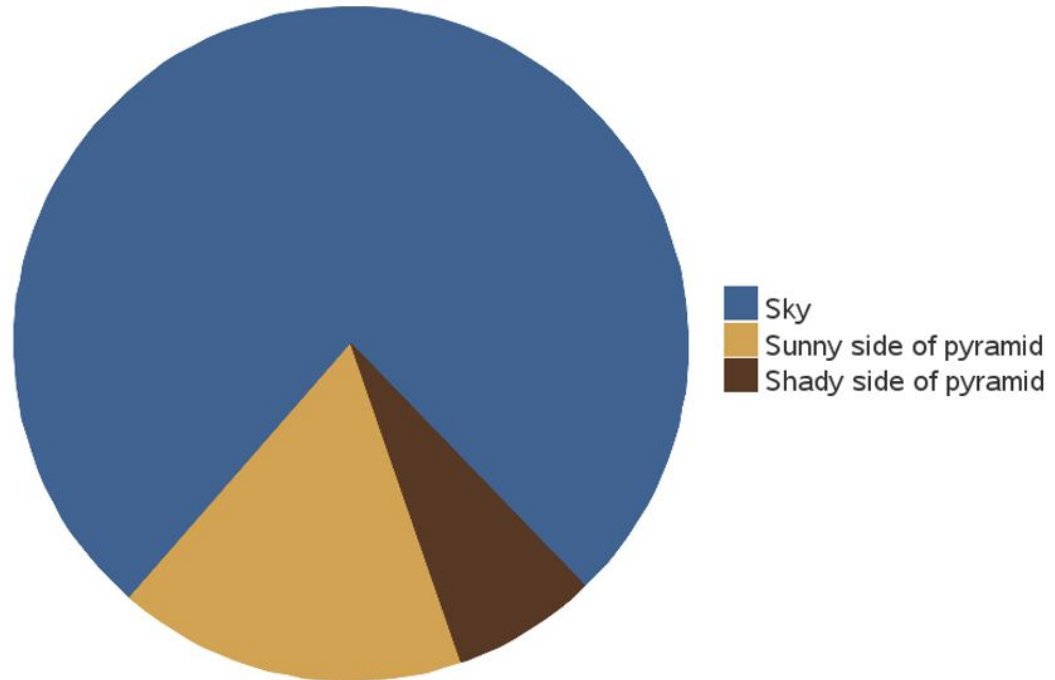
- Categorical (each slice), numerical (size)





When is a good time to use a pie chart?

- The short answer is only in very rare occasions.
- Ideally in interactive situations or if you're looking to see if exactly two things are equal. E.g.





Visualization Zoo

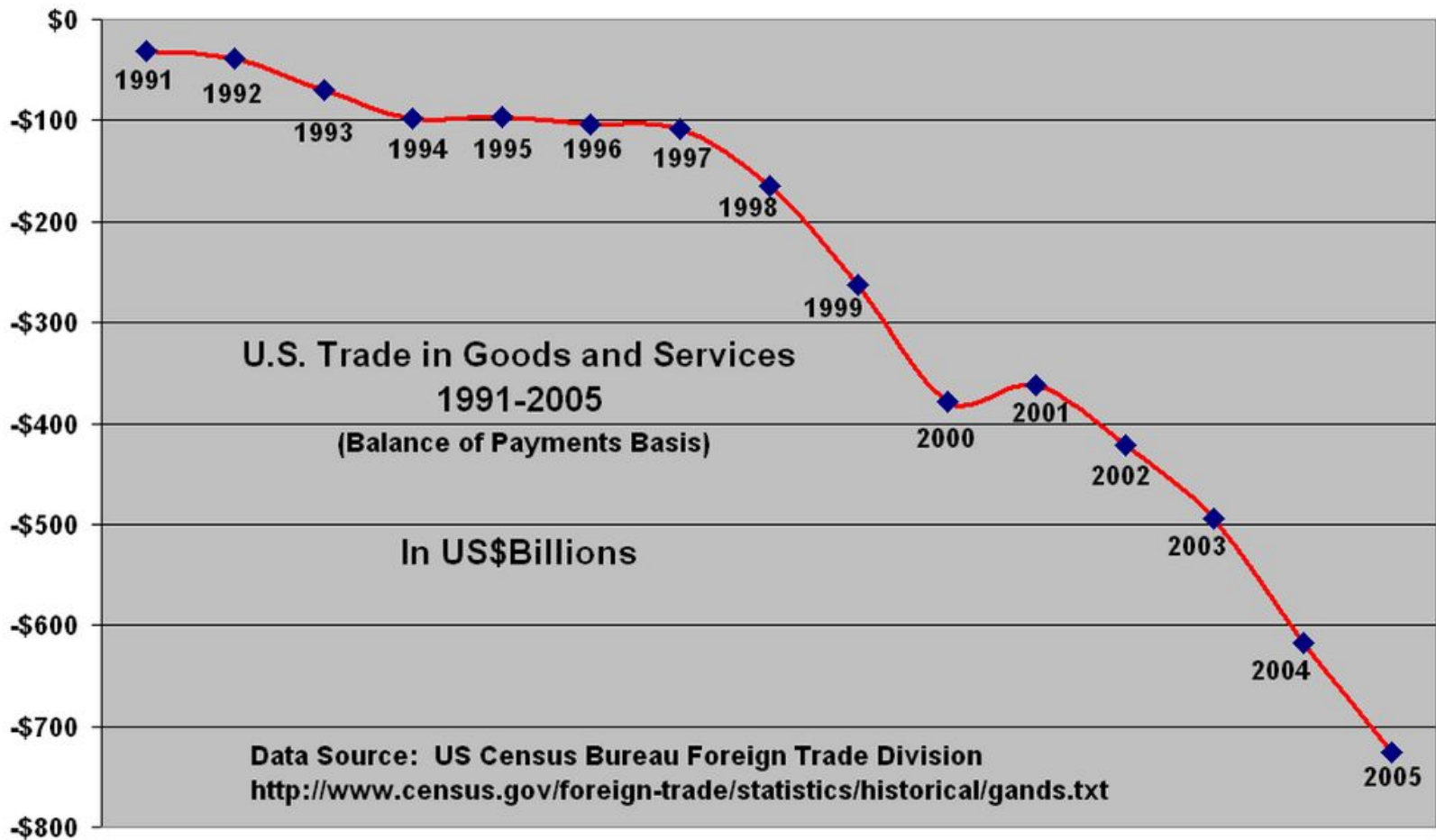
- Comparing Categories
 - Bar Chart: Bar, Stacked Bar, Grouped Bar
 - Pie Chart
- **Discovering Temporal Trends**
 - **Line, Multi-Line, Area Charts**
- Multiple Attributes
 - Radar Chart, Bubble Charts, Parallel Coordinates
- Visualizing Relationships and Hierarchy
 - Tree Maps, Scatterplots, Tree Diagrams, Network Graph



Line Chart/Graph

- Show trends over continuous intervals (e.g., months, years)
 - Good for spotting increases/decreases over time

Limitation(s): Doesn't work well for categorical or sparse time points





Multi-Line Chart

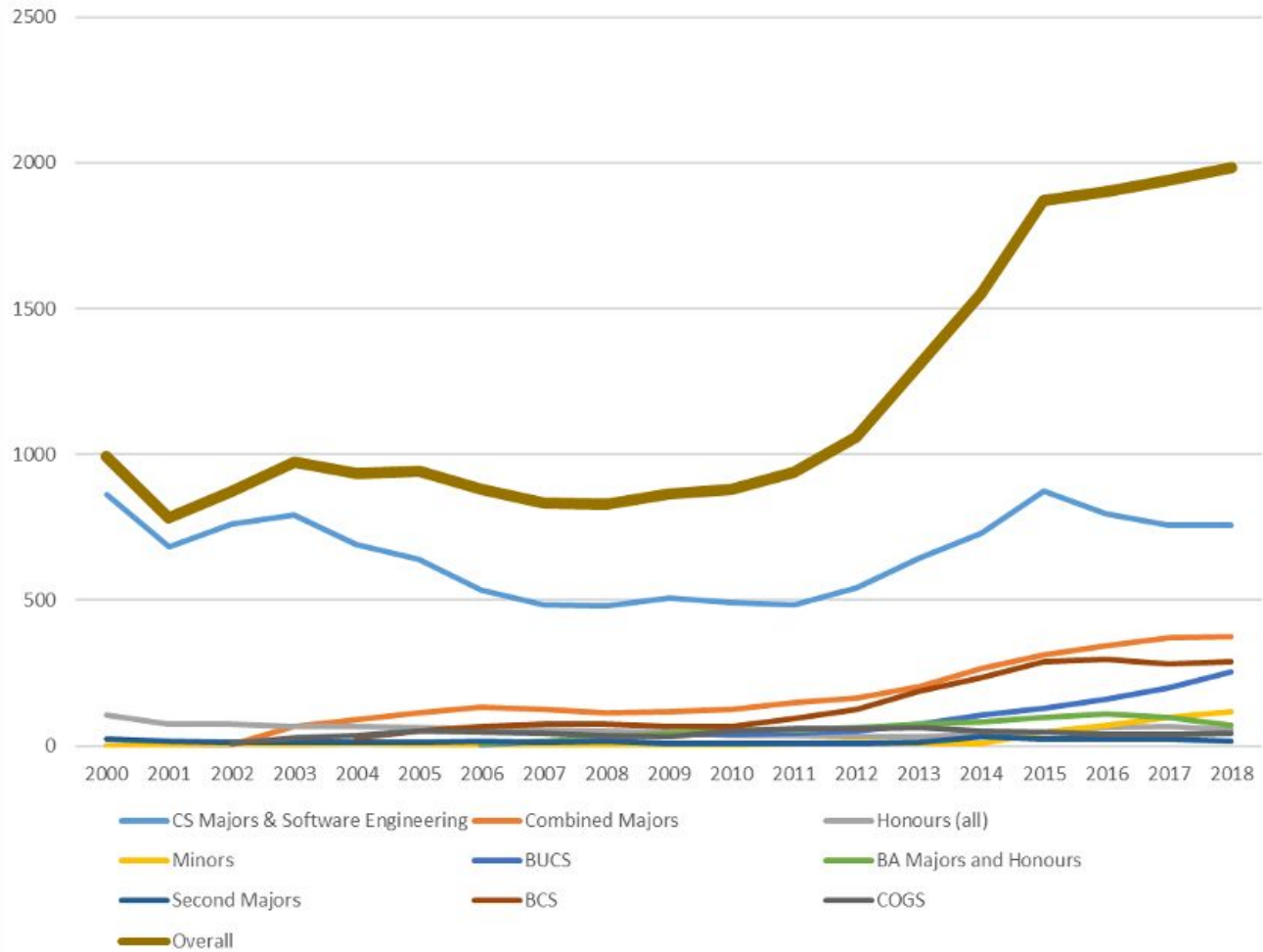
- Compare multiple trends simultaneously
 - Highlights similarities or differences across groups

Limitation(s):

- Colour/label clutter with more than 4–5 lines



Total Undergraduate Majors by Degree Type





Area Chart

- Show cumulative change or volume over time
 - Visual emphasis on totals and growth

Limitation(s): Overlapping areas can obscure individual trends

- Can be misleading if baseline isn't zero

Data type:

- Time series, numeric

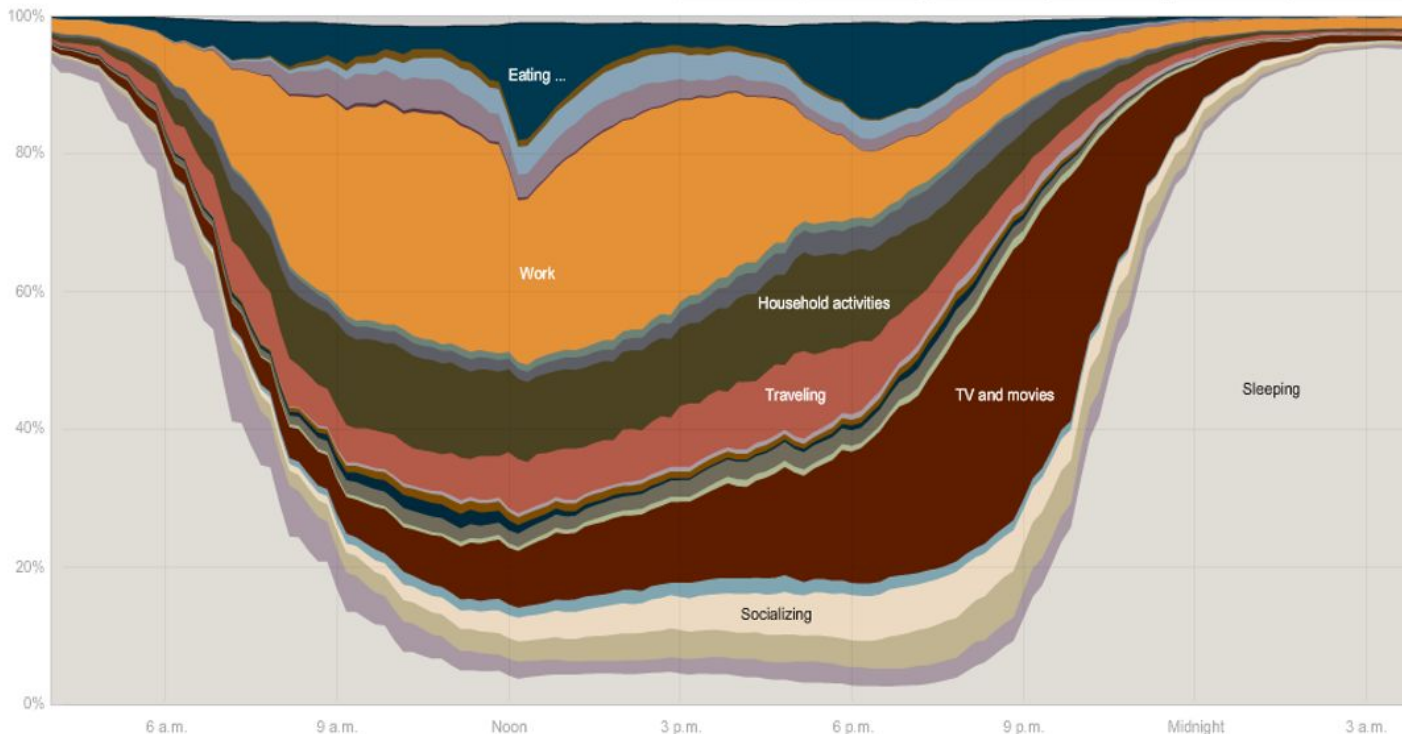
How Different Groups Spend Their Day

The American Time Use Survey asks thousands of American residents to recall every minute of a day. Here is how people over age 15 spent their time in 2008. [Related article](#)

Everyone

Sleeping, eating, working and watching television take up about two-thirds of the average day.

Everyone	Employed	White	Age 15-24	H.S. grads	No children
Men	Unemployed	Black	Age 25-64	Bachelor's	One child
Women	Not in lab...	Hispanic	Age 65+	Advanced	Two+ children







Visualization Zoo

- Comparing Categories
 - Bar Chart: Bar, Stacked Bar, Grouped Bar
 - Pie Chart
- Discovering Temporal Trends
 - Line, Multi-Line, Area Charts
- **Multiple Attributes**
 - **Radar Chart, Bubble Charts, Parallel Coordinates**
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Radar Chart (Spider Chart)

- Profile comparison across multiple attributes (e.g., performance metrics)
 - Great for showing shape or balance of characteristics

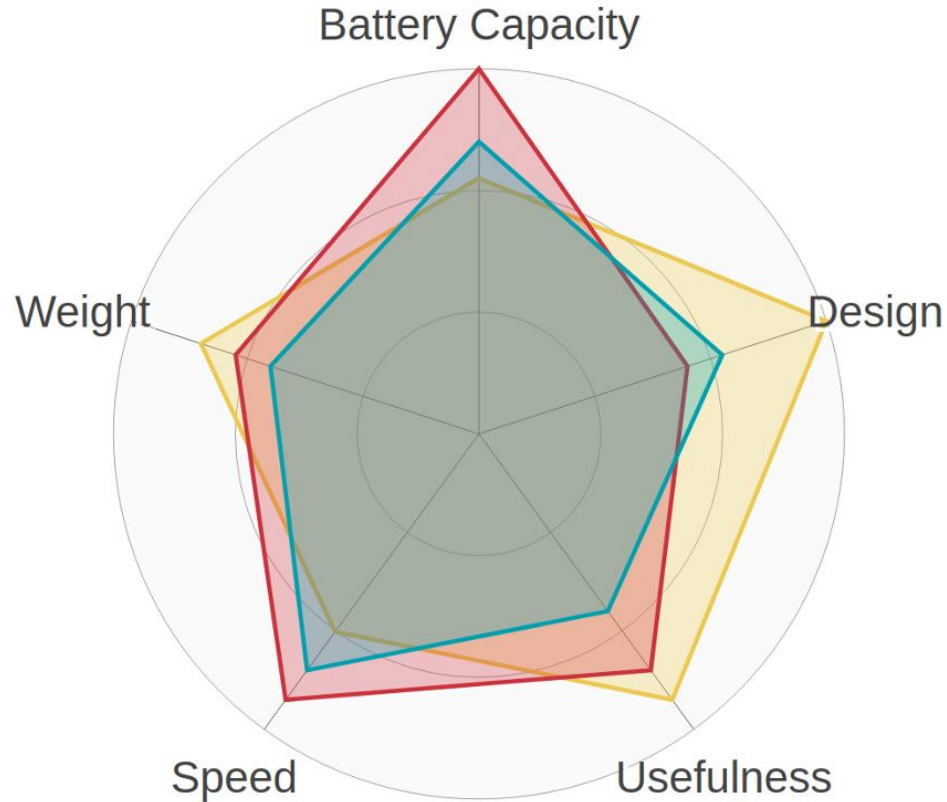
Limitation(s):

- Difficult to interpret beyond 5–7 axes
- Angles can distort perception

Data type:

- Multivariate (many numeric dimensions per category)

Radar Chart (Spider Chart)





Bubble Chart

- Show 3 variables in one chart (x, y, and bubble size)
 - Captures correlation and scale

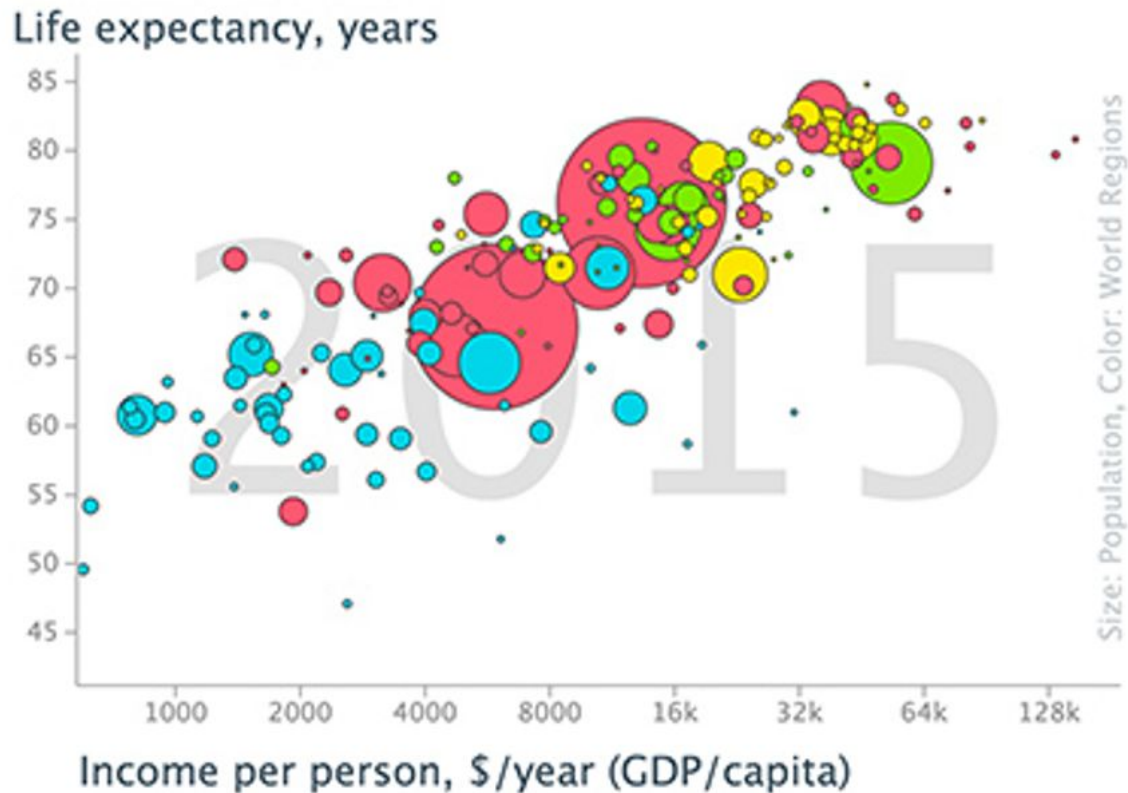
Limitation(s):

- Bubble size often misinterpreted (area vs. radius)
- Overlapping bubbles obscure meaning

Data type:

- Quantitative x, y, and magnitude

Bubble Chart





Parallel Coordinates

- Visualize high-dimensional numerical data
 - Detect patterns, clusters, and outliers

Limitation(s):

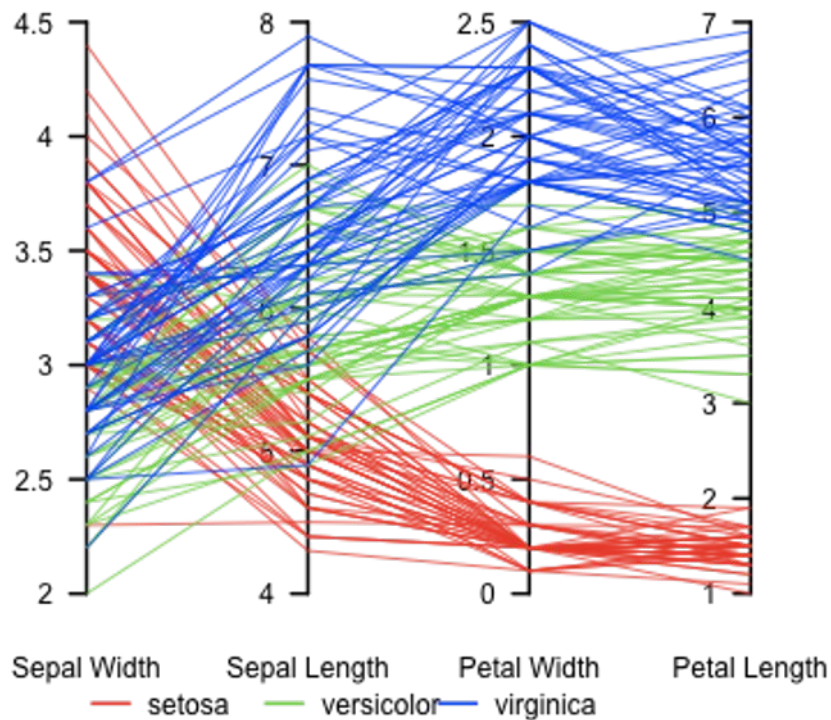
- Steep learning curve for novices
- Overplotting with many records

Data type:

- Multivariate numerical

Parallel Coordinates

Parallel coordinate plot, Fisher's Iris data





Visualization Zoo

- Comparing Categories
 - Bar Chart: Bar, Stacked Bar, Grouped Bar
 - Pie Chart
- Discovering Temporal Trends
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- Multiple Attributes
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- **Visualizing Relationships and Hierarchy**
 - **Tree Maps, Scatterplots, Tree Diagrams, Network Graph**



Tree Map

- Represent hierarchical parts-of-a-whole using nested rectangles
 - Space-efficient and good for proportional data

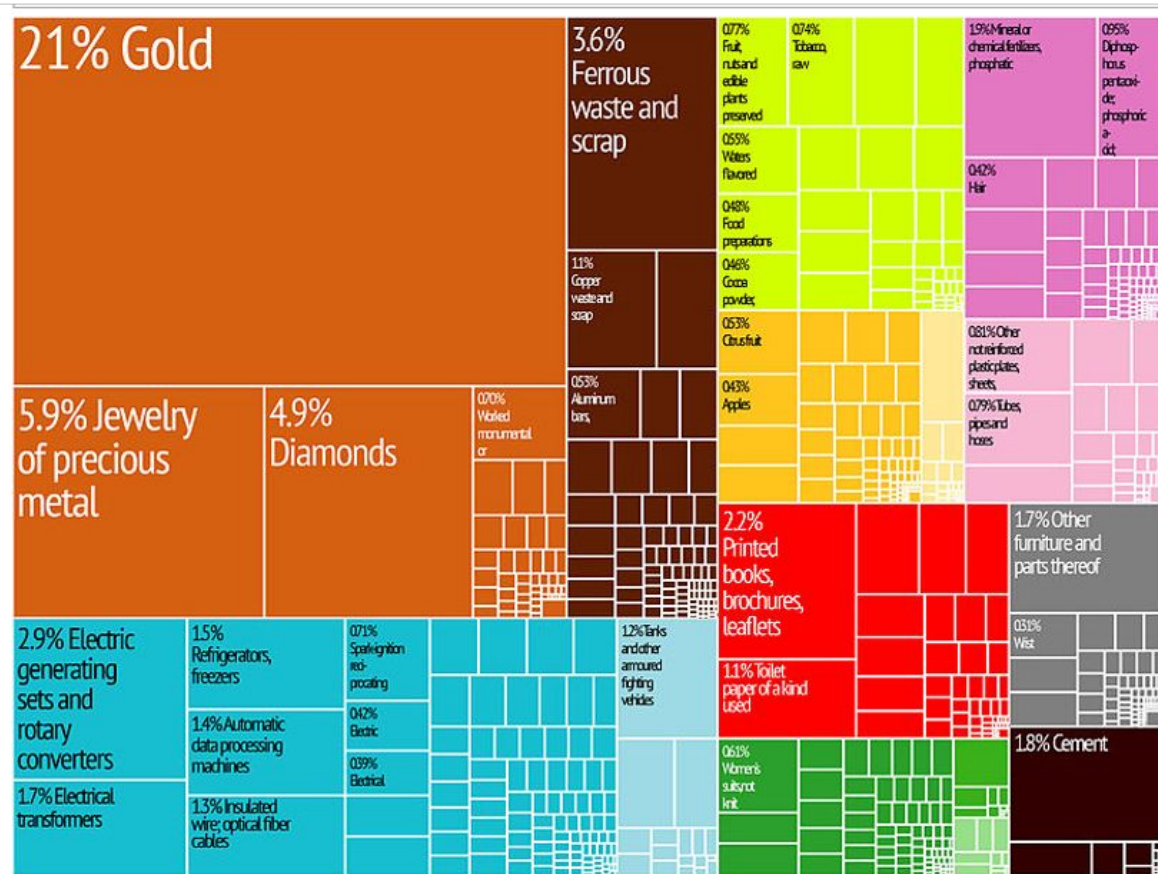
Limitation(s):

- Hard to compare small segments
- Poor for deep hierarchies

Data type:

- Hierarchical categorical + numerical

Tree Map





Scatterplot

- Show relationships between two quantitative variables
 - Reveals correlation, clusters, and outliers

Limitation(s):

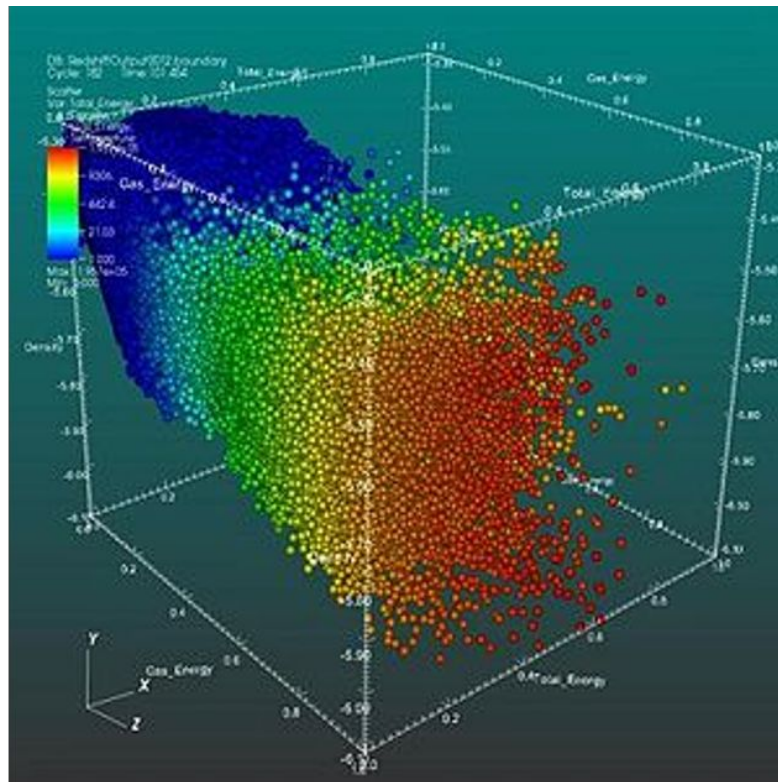
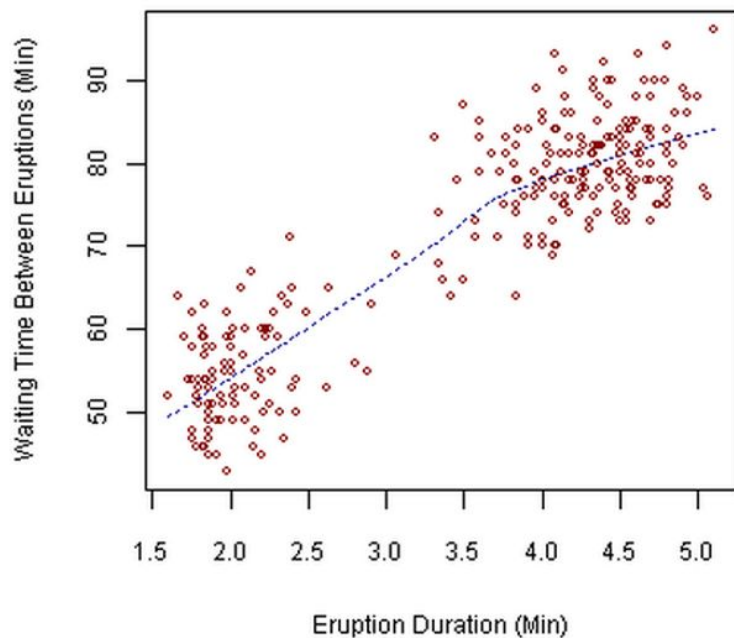
- Can't handle more than 2–3 variables without color/size

Data type:

- Quantitative x and y

Scatterplot

Old Faithful Eruptions





Network Graph

- Display relationships between entities (e.g., social networks, citations)
 - Highlights central nodes, connectivity

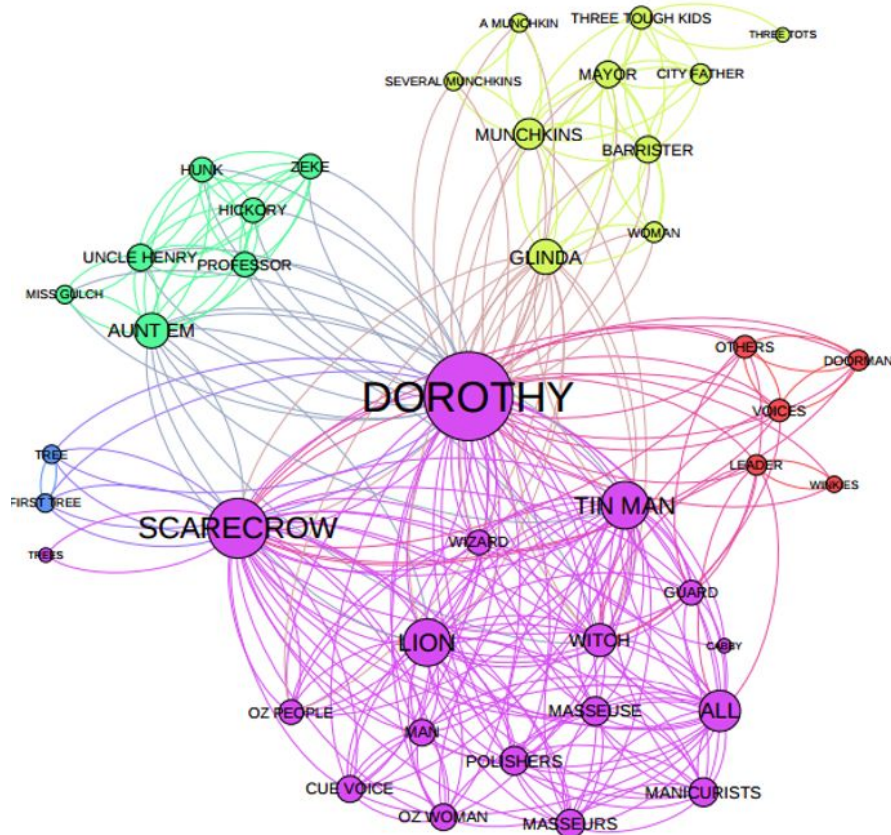
Limitation(s):

- Layout heavily affects readability
- Dense networks require interactivity

Data type:

- Nodes + edges (categorical and/or weighted)

Network Graph





Wrap up